```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, classification_report
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
import pandas as pd
df = pd.read_csv('student-mat (2).csv')
df.head()
```

	school;sex;age;address;famsize;Pstatus;Medu;Fedu;Mjob;Fjob;reason;guardian;traveltime;studytime;failures;schoolsup;famsup;pa
0	
1	
2	
3	
4	

```
df = pd.read_csv('student-mat (2).csv')
```

```
import os
print(os.listdir('/content'))
```

```
['.config', 'student-mat (2).csv', 'sample_data']
```

```
import pandas as pd
df = pd.read_csv('student-mat (2).csv', sep=';')
df.head()
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	ab
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	

5 rows × 33 columns

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    int64
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    object
 9   Fjob        395 non-null    object
 10  reason      395 non-null    object
 11  guardian    395 non-null    object
 12  traveltime  395 non-null    int64
 13  studytime   395 non-null    int64
 14  failures    395 non-null    int64
 15  schoolsup   395 non-null    object
 16  famsup      395 non-null    object
 17  paid        395 non-null    object
 18  activities  395 non-null    object
 19  nursery     395 non-null    object
 20  higher      395 non-null    object
 21  internet    395 non-null    object
 22  romantic    395 non-null    object
 23  famrel      395 non-null    int64
 24  freetime    395 non-null    int64
 25  goout       395 non-null    int64
 26  Dalc        395 non-null    int64
 27  Walc        395 non-null    int64
 28  health      395 non-null    int64
 29  absences    395 non-null    int64
 30  G1          395 non-null    int64
 31  G2          395 non-null    int64
 32  G3          395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443	3.108861	1.481013	2.
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862	1.113278	0.890741	1.
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000	2.000000	1.000000	1.
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000	3.000000	1.000000	2.
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000	4.000000	2.000000	3.
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000	5.000000	5.000000	5.

```
df.isnull().sum()
```

	0
school	0
sex	0
age	0
address	0
famsize	0
Pstatus	0
Medu	0
Fedu	0
Mjob	0
Fjob	0
reason	0
guardian	0
traveltime	0
studytime	0
failures	0
schoolsup	0
famsup	0
paid	0
activities	0
nursery	0
higher	0
internet	0
romantic	0
famrel	0
freetime	0
goout	0
Dalc	0
Walc	0
health	0
absences	0
G1	0
G2	0
G3	0

dtype: int64

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])
```

```
X = df.drop('G3', axis=1)
y = df['G3']
```

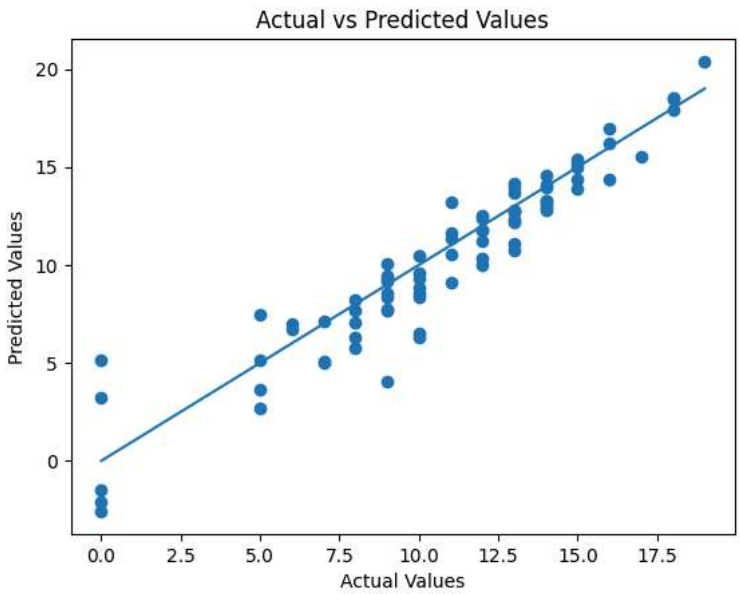
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv('student-mat (2).csv', sep=';')

le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])

X = df.drop('G3', axis=1)
y = df['G3']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

```
MSE: 2.384068062999674
R2: 0.8702362702448853
```

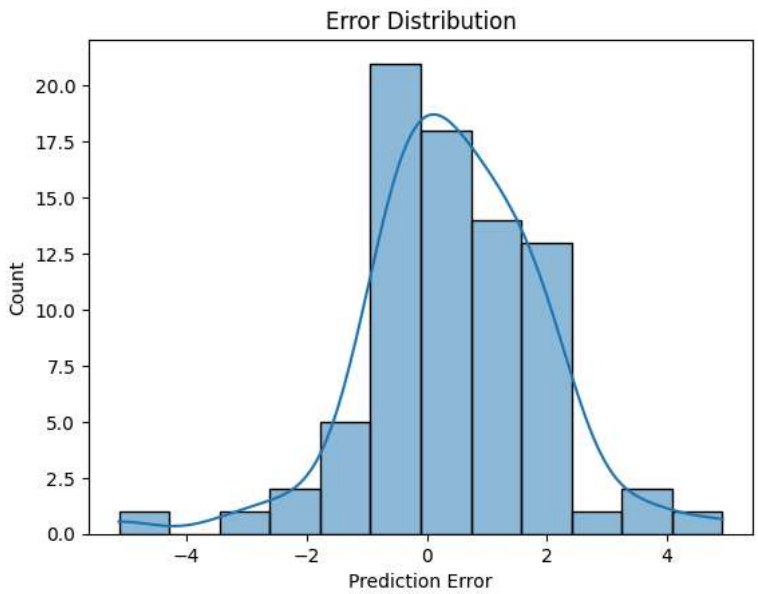
Actual vs Predicted Values

```
import matplotlib.pyplot as plt

# Example: actual and predicted values
y_test = y_test.values   # actual values
y_pred = model.predict(X_test)  # predicted values

# Plot
plt.figure()
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted Values")

# Line for perfect prediction
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()])

plt.show()
```



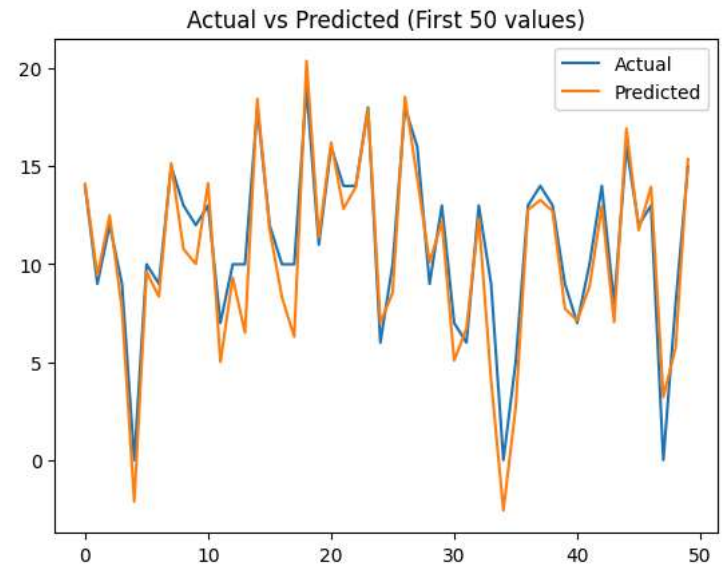
Error Distribution Graph

```
import seaborn as sns

errors = y_test - y_pred

plt.figure()
sns.histplot(errors, kde=True)
plt.title("Error Distribution")
plt.xlabel("Prediction Error")
plt.show()
```



Model Performance (Line Graph)

```
plt.figure()
plt.plot(y_test[:50], label="Actual")
plt.plot(y_pred[:50], label="Predicted")

plt.legend()
plt.title("Actual vs Predicted (First 50 values)")
plt.show()
```




```
from google.colab import files
uploaded = files.upload()
```

Choose Files student-mat (2).csv
student-mat (2).csv(text/csv) - 56993 bytes, last modified: 3/18/2026 - 100% done
Saving student-mat (2).csv to student-mat (2).csv

```
import pandas as pd

df = pd.read_csv("student-mat (2).csv")
```

```
print(df.columns)
```

```
time;failures;schoolsup;famsup;paid;activities;nursery;higher;internet;romantic;famrel;freetime;goout;Dalc;Walc;health;absences;G
```

```
df = pd.read_csv("student-mat (2).csv", sep=';')
```

```
print(df.columns)
```

```
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc',
       'Walc', 'health', 'absences', 'G1', 'G2', 'G3'],
      dtype='object')
```

```
X = df.drop("G3", axis=1)
y = df["G3"]
```

```
df = df.dropna(axis=1, how='all')
```

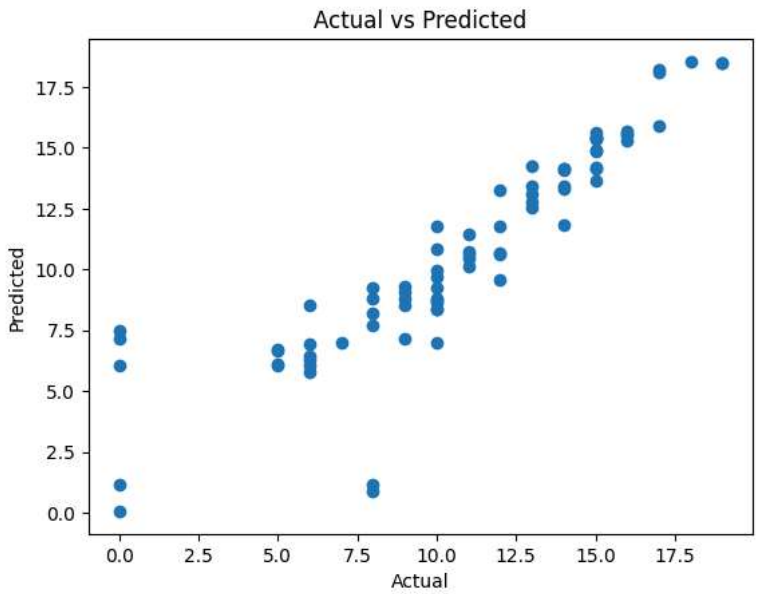
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset (IMPORTANT: sep=';')
df = pd.read_csv("student-mat (2).csv", sep=';')

# Convert categorical → numeric
df = pd.get_dummies(df, drop_first=True)

# Features & Target
```

```
X = df.drop("G3", axis=1)
y = df["G3"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
model = RandomForestRegressor()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MSE: 4.0133341772151905
R2 Score: 0.8042757900165661
```

```
import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")
plt.show()
```



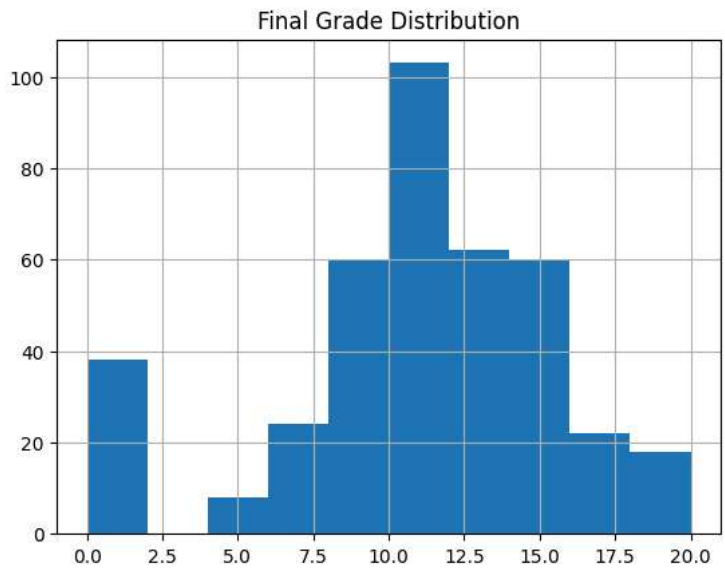

```
print(df.head())
print(df.info())
print(df.describe())
```

```
 37  activities_yes     395 non-null    bool
 38  nursery_yes        395 non-null    bool
 39  higher_yes         395 non-null    bool
 40  internet_yes       395 non-null    bool
 41  romantic_yes       395 non-null    bool
dtypes: bool(26), int64(16)
memory usage: 59.5 KB
None
              age        Medu        Fedu  traveltime   studytime    failures  \
count  395.000000  395.000000  395.000000  395.000000  395.000000  395.000000
mean    16.696203    2.749367    2.521519    1.448101    2.035443    0.334177
std      1.276043    1.094735    1.088201    0.697505    0.839240    0.743651
min     15.000000    0.000000    0.000000    1.000000    1.000000    0.000000
25%     16.000000    2.000000    2.000000    1.000000    1.000000    0.000000
50%     17.000000    3.000000    2.000000    1.000000    2.000000    0.000000
75%     18.000000    4.000000    3.000000    2.000000    2.000000    0.000000
max     22.000000    4.000000    4.000000    4.000000    4.000000    3.000000

           famrel    freetime       goout        Dalc        Walc      health  \
count  395.000000  395.000000  395.000000  395.000000  395.000000  395.000000
mean     3.944304    3.235443    3.108861    1.481013    2.291139    3.554430
std      0.896659    0.998862    1.113278    0.890741    1.287897    1.390303
min      1.000000    1.000000    1.000000    1.000000    1.000000    1.000000
25%      4.000000    3.000000    2.000000    1.000000    1.000000    3.000000
50%      4.000000    3.000000    3.000000    1.000000    2.000000    4.000000
75%      5.000000    4.000000    4.000000    2.000000    3.000000    5.000000
max      5.000000    5.000000    5.000000    5.000000    5.000000    5.000000

         absences          G1          G2          G3
count  395.000000  395.000000  395.000000  395.000000
mean     5.708861   10.908861   10.713924   10.415190
std      8.003096    3.319195    3.761505    4.581443
min      0.000000    3.000000    0.000000    0.000000
25%      0.000000    8.000000    9.000000    8.000000
50%      4.000000   11.000000   11.000000   11.000000
75%      8.000000   13.000000   13.000000   14.000000
max     75.000000   19.000000   19.000000   20.000000
```

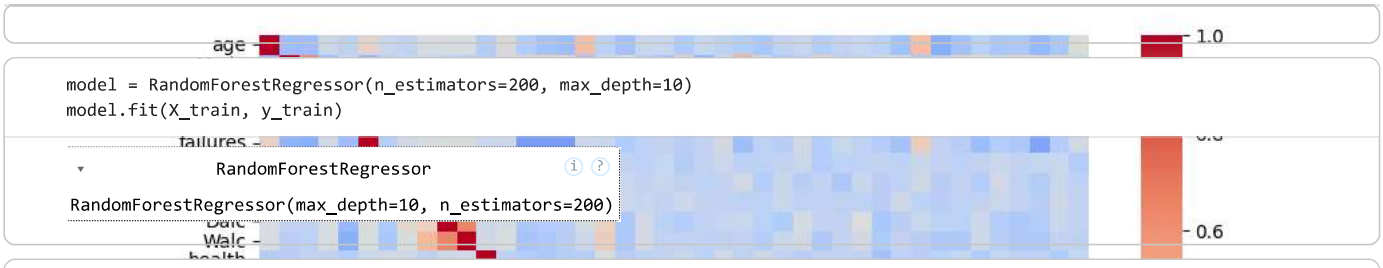
```
print(df.isnull().sum())
```

```
age                 0
Medu                0
Fedu                0
traveltime          0
studytime           0
failures            0
famrel              0
freetime            0
goout               0
Dalc                0
Walc                0
health              0
absences            0
G1                  0
G2                  0
G3                  0
school_MS           0
sex_M               0
address_U           0
famsize_LE3         0
Pstatus_T           0
Mjob_health         0
Mjob_other          0
Mjob_services       0
Mjob_teacher        0
Fjob_health         0
Fjob_other          0
Fjob_services       0
Fjob_teacher        0
reason_home         0
reason_other        0
reason_reputation   0
guardian_mother     0
guardian_other      0
schoolsup_yes       0
famsup_yes          0
paid_yes            0
activities_yes      0
nursery_yes         0
higher_yes          0
internet_yes        0
romantic_yes        0
dtype: int64
```

```
import matplotlib.pyplot as plt

# Histogram
df['G3'].hist()
plt.title("Final Grade Distribution")
plt.show()
```



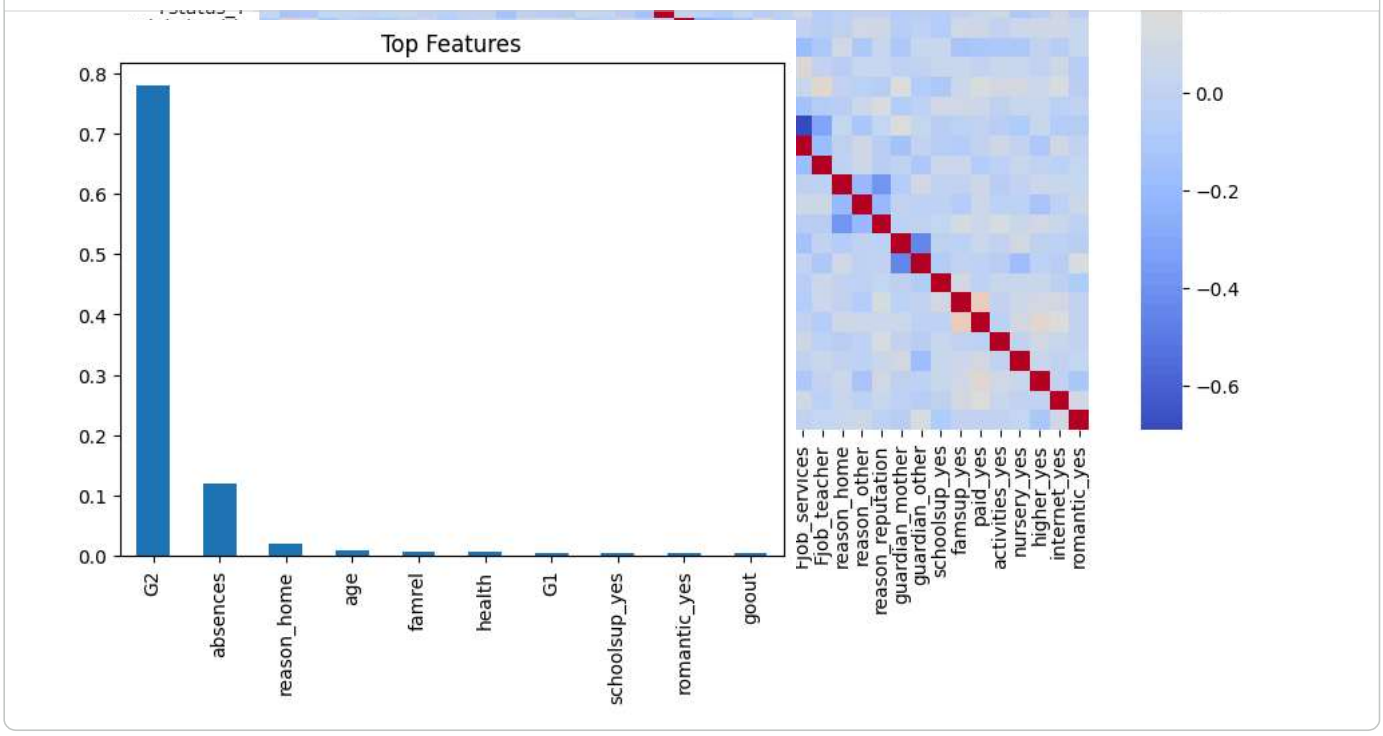

```
import seaborn as sns

plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), cmap='coolwarm')
plt.show()
```




```
model = RandomForestRegressor(n_estimators=200, max_depth=10)
model.fit(X_train, y_train)
```

```
RandomForestRegressor
RandomForestRegressor(max_depth=10, n_estimators=200)
```

```
import pandas as pd

importance = pd.Series(model.feature_importances_, index=X.columns)
importance.sort_values(ascending=False).head(10).plot(kind='bar')
plt.title("Top Features")
plt.show()
```




```
print("MSE:", mean_squared_error(y_test, y_pred))
```